

International Information Technology University

UDC: 004.93:617.735

On manuscript right

YESMUKHAMEDOV NURMAGANBET SEITKALIULY

**Automated Diabetic Retinopathy Diagnosis via Fundus Image
Enhancement and CNN Classification**

8D06104 – Computer systems and software engineering

Thesis for the degree of doctor of
Philosophy (PhD)

Scientific consultant
Candidate of Phys.-Math. Sciences,
Associate Professor, International
Information Technology University
Sapakova S.Z.

Foreign consultant
Professor, Universiti Putra Malaysia
Al-Haddad S.A.R.

Republic of Kazakhstan
Almaty, 2026

CONTENT

NORMATIVE REFERENCES.....	—
DESIGNATIONS AND ABBREVIATIONS.....	—
DEFINITIONS	—
INTRODUCTION	1
1 PROBLEM DOMAIN ANALYSIS AND CURRENT STATE OF AUTOMATED DIABETIC RETINOPATHY DIAGNOSIS	21
1.1 Medical and Epidemiological Context of Diabetic Retinopathy	21
1.1.1 Pathophysiology and Clinical Grading Systems	21
1.1.2 Screening Requirements in Resource-Limited Healthcare Settings	24
1.2 Fundus Image Acquisition and Quality Variability	26
1.2.1 Sources of Image Degradation in Clinical Practice	26
1.2.2 Impact of Image Quality on Diagnostic Model Performance	28
1.2.3 Device-Specific Variability in Fundus Imaging.....	31
1.3 Deep Learning Approaches to Retinal Image Classification.....	33
1.3.1 CNN Architectures for Medical Imaging.....	33
1.3.2 Transfer Learning and Self-Supervised Pretraining in Ophthalmic Diagnostics	36
1.3.3 Explainability Methods in Medical Image Classification.....	38
1.4 Critical Analysis of Existing Automated DR Screening Systems	40
1.5 Formulation of the Research Problem	44
Conclusions to Chapter 1.....	46
2 THEORETICAL FOUNDATIONS OF IMAGE PREPROCESSING AND DEEP LEARNING FOR FUNDUS IMAGE ANALYSIS	47
2.1 Mathematical Foundations of Image Enhancement Techniques	47
2.1.1 Histogram Equalization and Adaptive Contrast Enhancement.....	47
2.1.2 Formalization of CLAHE with Dual-Constraint Clip Limit.....	50
2.1.3 Spatial Filtering and Noise Reduction Methods	54
2.2 Theoretical Framework of Convolutional Neural Networks	56
2.2.1 Convolution, Pooling, and Feature Extraction Operations	56
2.2.2 Loss Functions and Optimization for Imbalanced Medical Datasets	58
2.2.3 Regularization Techniques: Dropout, Batch Normalization, and Data Augmentation	60
2.3 Transfer Learning and Self-Supervised Representation Learning Theory	62
2.3.1 Feature Transferability Across Visual Domains	62
2.3.2 Frozen-Layer versus Progressive Fine-Tuning Strategies	64
2.3.3 In-Domain Self-Supervised Pretraining for Retinal Imaging	66
2.4 Mathematical Modeling of Laser-Tissue Interaction in Retinal Therapy.....	68
2.4.1 Coupled Thermal-Optical Model of Fundus Tissue Response	68
2.5 Explainability in Deep Learning for Medical Imaging.....	146

2.5.1 CAM / Grad-CAM Theory and Formalization	70
2.5.2 Attention Map Interpretation.....	72
2.5.3 ALO and IoU as Quantitative Explainability Metrics.....	74
2.6 Image Quality Metrics for Preprocessing Evaluation: CNR, VVI, Entropy, SSIM.....	75
Conclusions to Chapter 2.....	78
3 METHODOLOGY OF INTEGRATED PREPROCESSING-CNN PIPELINE DESIGN.....	79
3.1 Formalization of the Unified Preprocessing Pipeline	79
3.1.1 Pipeline Stage Specification: 8-Stage System.....	79
3.1.2 Modified CLAHE Algorithm with Dual-Constraint Clip Limit	95
3.1.3 Augmentation Strategy for Class Imbalance Mitigation.....	98
3.1.4 External Image Ingestion Protocol	102
3.2 Design of CNN Architectures for DR Classification.....	103
3.2.1 ResNet-50 and EfficientNet-B3 as Primary Experimental Architectures..	103
3.2.2 Historical Reference Architectures (Reference Only)	108
3.3 Transfer Learning and Pretraining Methodology.....	109
3.3.1 Architecture Adaptation for Five-Class DR Classification.....	109
3.3.2 Ophthalmology-Specific Self-Supervised Pretraining of the CNN Backbone (Integrated Arm).....	110
3.3.3 Two-Stage Fine-Tuning Protocol Design.....	113
3.3.4 Weighted Loss Function Formulation for Ordinal Class Structure	114
3.4 Evaluation Framework and Performance Metrics	116
3.4.1 Multi-Metric Assessment Framework.....	116
3.4.2 Cross-Validation and Statistical Reliability Protocols	119
Conclusions to Chapter 3.....	121
4 EXPERIMENTAL RESEARCH — PREPROCESSING IMPACT ON CNN DIAGNOSTIC PERFORMANCE	123
4.1 Datasets and Experimental Configuration	123
4.1.1 Dataset Architecture	123
4.1.2 Class Distribution and Data Partitioning.....	128
4.1.3 Hardware and Reproducibility Protocol.....	130
4.2 Experiment 1: Integrated Pipeline Dominance — Pipeline + In-Domain Pretraining vs. Baseline on EyePACS (H-1)	132
4.2.1 Restored 2×2 Factorial Design (Configurations A–D)	132
4.2.2 Training Dynamics and Convergence Analysis	134
4.2.3 Quantitative Comparison of Diagnostic Metrics.....	137
4.3 Experiment 2: Pipeline Stage Ablation + CLAHE/ σ Sweeps (H-2).....	142
4.3.1 Cumulative Ablation Design (Levels L0–L7).....	142
4.3.2 CLAHE Threshold Sensitivity Analysis (H-2 Sub-Analysis).....	146
4.3.3 Flat-Field σ Sweep and Image Quality Metrics	149

4.4 Domain Distance Reduction Across Six Target Domains (H-3).....	152
4.4.1 Measurement Protocol: MMD over Representations and KL over Channel Histograms.....	152
4.4.2 Distance Reduction Results and Their Interpretive Limits	154
4.5 Experiment 3: Cross-Dataset Transferability on APTOS 2019 (H-4).....	158
4.5.1 Zero-Shot Transfer to APTOS 2019	158
4.5.2 Baseline vs Pipeline Comparison	160
4.6 Experiment 4: Grad-CAM Explainability on IDRiD + Clinical (H-5).....	161
4.6.1 Grad-CAM Generation Protocol	161
4.6.2 Quantitative ALO and IoU with IDRiD Lesion Masks	163
4.6.3 Per-Image Consistency of the Attention Effect and Limits of the Present Evidence	166
4.7 Experiment 5: External Clinical Performance on IDRiD + Messidor-2 (H-7)...	167
4.8 Experiment 6: Device Domain Shift on DDR + ODIR-5K + RFMiD (H-6)	170
4.9 Experiment 7: Small Data Training (IDRiD → Clinical).....	172
Conclusions to Chapter 4.....	174
5 RELIABILITY VALIDATION AND COMPARATIVE ANALYSIS	177
5.1 Explainability Results	177
5.2 Statistical Validation	180
5.2.1 Bootstrap Confidence Intervals and Mixed-Effects Model	180
5.2.2 Final Claim Strength Classifications	183
5.3 Comparative Analysis with Published Systems.....	187
5.3.1 Benchmarking Against Published Results: IDx-DR, EyeNuk, DeepMind	187
5.3.2 Performance–Complexity Trade-Off Analysis	191
5.4 Limitations and Boundary Conditions of the Proposed Approach	193
Conclusions to Chapter 5.....	196
6 ARCHITECTURE OF AN AUTOMATED DR SCREENING SYSTEM FOR RESOURCE-LIMITED ENVIRONMENTS	198
6.1 System Requirements and Design Principles.....	198
6.1.1 Functional and Non-Functional Requirements Specification	198
6.1.2 Modular Architecture with PACS and EHR Integration.....	201
6.2 AI Processing Module Design	206
6.2.1 Preprocessing Engine with Configurable Pipeline.....	206
6.2.2 Inference Module with Model Selection Logic.....	208
6.3 Clinical Workflow Integration	210
6.3.1 Telemedicine and Portable Device Support for Rural Deployment.....	210
6.3.1.1 Deployment in distributed telemedicine systems	211
6.3.1.2 Integration with national eHealth platforms	211
6.3.1.3 Real-time remote DR screening in low-resource regions.....	212
6.3.2 Physician-in-the-Loop Decision Support Interface.....	213

6.4 Data Security and Regulatory Compliance Framework 214

6.4.1 GDPR/HIPAA-Aligned Data Management Protocols 214

6.4.2 Applicability to Kazakhstan Healthcare Infrastructure..... 216

Conclusions to Chapter 6..... 217

CONCLUSION..... 218

LIST OF REFERENCES USED 221

APPENDIX A 230

APPENDIX B..... 234

APPENDIX C 239

APPENDIX D 244

APPENDIX E..... 252

APPENDIX F..... 267

NORMATIVE REFERENCES

This thesis uses references to the following standards:

Instructions for the preparation of a thesis and an abstract, Higher Attestation Commission of the Ministry of Education and Science of the Republic of Kazakhstan dated September 28, 2004 No. 377-3y.

GOST 7.32-2001. Report on research work. Structure and design rules.

GOST 7.1-2003. Bibliographic record. Bibliographic description. General requirements and rules of compilation.

ST RK 34.005-2002. Information Technology. Basic terms and definitions (first edition).

ST RK 34.015-2002. Information Technology. Set of standards for automated systems. Terms of reference for creating an IS (first edition).

ST RK 34.027-2006. Information Technologies. Classification of software tools (first edition).

ST RK 34.014-2002. Information Technology. A set of standards for automated systems. Automated systems. Terms and definitions.

DESIGNATIONS AND ABBREVIATIONS

AI	Artificial Intelligence
ALO	Attention–Lesion Overlap (explainability metric)
APTOS	Asia Pacific Tele-Ophthalmology Society (2019 Blindness Detection dataset)
AUC	Area Under the Curve
BYOL	Bootstrap Your Own Latent (self-supervised learning method)
CAM	Class Activation Mapping
CLAHE	Contrast-Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
CNR	Contrast-to-Noise Ratio
DDR	Diabetic Retinopathy grading dataset (DDR)
DINO	self-Distillation with No labels (self-supervised learning method)
DR	Diabetic Retinopathy
ECE	Expected Calibration Error
EHR	Electronic Health Record
EMD	Earth Mover's Distance
EyePACS	Eye Picture Archive Communication System (diabetic-retinopathy dataset)
FOV	Field of View
GDPR	General Data Protection Regulation
Grad-CAM	Gradient-weighted Class Activation Mapping
HIPAA	Health Insurance Portability and Accountability Act
IDRiD	Indian Diabetic Retinopathy Image Dataset
IoU	Intersection over Union
KL	Kullback–Leibler divergence
LAB	CIELAB colour space
MLP	Multilayer Perceptron
MoCo	Momentum Contrast (self-supervised learning method)
NPV	Negative Predictive Value
OD	Optic Disc
ODIR-5K	Ocular Disease Intelligent Recognition dataset (5,000 patients)
PACS	Picture Archiving and Communication System
PPV	Positive Predictive Value
ReLU	Rectified Linear Unit
RFMiD	Retinal Fundus Multi-disease Image Dataset
RIQA	Retinal Image Quality Assessment
ROC	Receiver Operating Characteristic
SimCLR	Simple framework for Contrastive Learning of Representations
SSIM	Structural Similarity Index
SSL	Self-Supervised Learning
STARE	Structured Analysis of the Retina (dataset)
UML	Unified Modeling Language
VRAM	Video Random-Access Memory

VVI	Vessel Visibility Index
WSL	Windows Subsystem for Linux
XAI	Explainable Artificial Intelligence
D	Field-of-view diameter, in pixels
CL	Clip limit of the CLAHE transform
F1	F1-score (harmonic mean of precision and recall)
G	Generalization ratio, $G = F1_external / F1_EyePACS$
T	Global threshold of the dual-constraint CLAHE (T/80 formulation)
α	Class-weight parameter of the Focal Loss (inverse-frequency)
γ	Focusing parameter of the Focal Loss ($\gamma = 2$)
κ	Cohen's Kappa with quadratic weights
σ	Standard deviation of the Gaussian kernel in flat-field correction, $\sigma = 0.07 \cdot D$
$\Delta F1$	Cross-dataset F1 drop, $\Delta F1 = F1_EyePACS - F1_external$

DEFINITIONS

Architectural complexity — the capacity of a CNN as defined by the number of convolutional layers, total trainable-parameter count, filter-size range, and the presence or absence of regularization components (batch normalization, dropout).

Baseline (baseline arm) — the reference configuration of Experiment 1: stretch-resize to 512×512 followed by ImageNet normalization, producing a 3-channel tensor with no field-of-view mask and no preprocessing stages; the backbone is initialized from ImageNet.

CLAHE (Contrast-Limited Adaptive Histogram Equalization) — an adaptive contrast-enhancement method operating on local image tiles with a clip limit that bounds contrast amplification; applied in the pipeline to the LAB L-channel under a dual-constraint clip limit (Stage 5).

Composite independent variable — the combined manipulated factor of Experiment 1 (H-1), in which the full-pipeline arm differs from baseline jointly along the preprocessing axis (eight-stage pipeline vs. stretch-resize + ImageNet normalize) and the pretraining axis (ophthalmology-specific self-supervised pretraining vs. ImageNet).

Convolutional Neural Network (CNN) — a neural architecture comprising convolutional and pooling layers for feature extraction and fully connected layers for classification; the classification backbone operating on preprocessed fundus images for five-class DR staging.

Cross-validation — a validation strategy using 5-fold patient-level stratified splitting; for each fold, four folds serve as training data and one as test data, with no patient's images appearing in both partitions; metrics are reported as mean $\pm$ standard deviation across folds.

Data augmentation — image transformations (unified affine, ColorJitter [brightness/contrast/saturation/hue], Gaussian noise, JPEG compression) applied to the training data to increase variability and improve generalization; the train-only Stage 6 of the pipeline.

Dataset-specific normalization — channel-wise normalization (Stage 7) using mean and standard deviation computed from the training split of the primary dataset rather than ImageNet statistics; the normalization of the full pipeline.

Diabetic retinopathy (DR) — a microvascular complication of diabetes mellitus and a leading cause of preventable vision loss, graded in five severity stages (0 — none, 1 — mild, 2 — moderate, 3 — severe, 4 — proliferative).

Diagnostic effectiveness — the joint performance profile on four primary metrics (accuracy, weighted F1-score, ROC-AUC, and Cohen's Kappa with quadratic weights) computed on the held-out test partition.

Domain shift — the distributional difference between training data and target deployment data arising from differences in imaging equipment, patient populations, or acquisition protocols.

Field-of-view (FOV) mask — a binary spatial mask (1.0 = real fundus data, 0.0 = zero-padding) generated at Stage 3 and appended as the 4th input channel, allowing the CNN to distinguish genuine retinal pixels from padded background.

Flat-field correction (adaptive) — Gaussian-blur subtraction with adaptive $\sigma = 0.07 \cdot D$ (D = FOV diameter), applied inside the FOV mask to normalize uneven illumination while preserving local contrast; Stage 4 of the pipeline.

Fundus image — a retinal photograph acquired by fundoscopy; the exclusive imaging modality of the dissertation and the input to the preprocessing pipeline and the CNN classifier.

Generalization (cross-database) — the ratio of test-set F1-score on a secondary dataset to test-set F1-score on the primary dataset under the same trained model without retraining; the generalization ratio $G = F1_external / F1_EyePACS$.

Grad-CAM (Gradient-weighted Class Activation Mapping) — an explainability method producing class-discriminative localization maps from a gradient-weighted combination of final convolutional feature maps; an interpretability tool, not a clinical localization of pathology.

Image quality — the measurable capacity of a fundus image to support automated detection of microvascular features relevant to DR staging, assessed through downstream classification performance rather than a subjective visual score.

Integrated pipeline dominance (H-1) — the primary hypothesis that the integrated full-pipeline configuration outperforms the baseline as a unitary system, validated when the empirical dominance criterion (≥ 5 pp weighted-F1 gain, ≥ 0.02 ROC-AUC gain, no Cohen's Kappa degradation) is satisfied.

Ophthalmology-specific self-supervised pretraining — self-supervised pretraining of the CNN backbone on an unlabeled retinal fundus corpus (no DR labels), used as the full-pipeline-arm initialization; learns fundus-domain representations directly on the 4-channel pipeline tensor.

Physician-in-the-loop — a design paradigm in which the clinician acts not merely as a consumer of AI outputs but as a tuner and auditor who interprets results and adjusts the system; the proposed system is a decision-support tool within this paradigm.

Preprocessing pipeline — the canonical ordered eight-stage sequence (Stages 0–7) applied to fundus images prior to CNN input; the full pipeline applies all eight stages, producing a 4-channel tensor (RGB + FOV mask).

Resource-limited environment — a deployment context characterized by at least two of: absence of GPU acceleration; available RAM below 16 GB; batch-processing-time constraints; network connectivity precluding continuous cloud-API reliance.

Transfer learning — the reuse and adaptation of pretrained network weights to the fundus-image domain; under H-1 the initialization source is slaved to the arm (ImageNet for baseline, ophthalmology-specific SSL for the full pipeline).

Weighted loss function — a cross-entropy loss with class-specific weights addressing class imbalance and exploiting the ordinal structure of the five-class DR grading.